

Electron current is obtained due to the movement of free electrons, whereas, Conventional current is obtained due to the movement of free positive charges. Both of these are called as electric current.

Voltage

The voltage "V" is nothing but an electromotive force that causes the charge (electrons) to flow. Mathematically, it can be written as

$$V = \frac{dW}{dQ}$$

Where,

- W is the potential energy and its unit is Joule.
- Q is the charge and its unit is Coloumb.

As an analogy, Voltage can be thought of as the pressure of water that causes the water to flow through a pipe. It is measured in terms of Volt.

Power

The power "P" is nothing but the time rate of flow of electrical energy. Mathematically, it can be written as

$$P = \frac{dW}{dt}$$

Where,

- W is the electrical energy and it is measured in terms of Joule.
- t is the time and it is measured in seconds.

We can re-write the above equation as

$$P = \frac{dW}{dt} = \frac{dW}{dQ} \times \frac{dQ}{dt} = VI$$

Therefore, power is nothing but the product of voltage V and current I. Its unit is Watt.

Types of Network Elements

We can classify the Network elements into various types based on some parameters.

Following are the types of Network elements –

- Active Elements and Passive Elements
- Linear Elements and Non-linear Elements
- Bilateral Elements and Unilateral Elements
- Lumped Elements and Distributed Elements

Active Elements and Passive Elements

We can classify the Network elements into either active or passive based on the ability of delivering power.

- Active Elements deliver power to other elements, which are present in an electric circuit. Sometimes, they may absorb the power like passive elements. That means active elements have the capability of both delivering and absorbing power.

Examples: Voltage sources and current sources.

- Passive Elements can't deliver power (energy) to other elements, however they can absorb power. That means these elements either dissipate power in the form of heat or store energy in the form of either magnetic field or electric field.

Examples: Resistors, Inductors, and capacitors.

Linear Elements and Non-Linear Elements

We can classify the network elements as linear or non-linear based on their characteristic to obey the property of linearity.

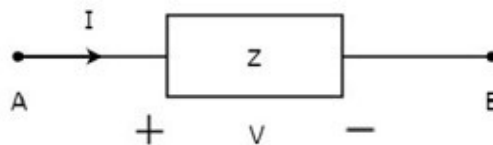
- Linear Elements are the elements that show a linear relationship between voltage and current. Examples: Resistors, Inductors, and capacitors.
- Non-Linear Elements are those that do not show a linear relation between voltage and current. Examples: Voltage sources and current sources.

Bilateral Elements and Unilateral Elements

Network elements can also be classified as either bilateral or unilateral based on the direction of current flows through the network elements.

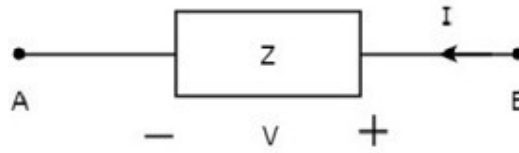
Bilateral Elements are the elements that allow the current in both directions and offer the same impedance in either direction of current flow. Examples: Resistors, Inductors and capacitors.

The concept of Bilateral elements is illustrated in the following figures.



In the above figure, the current (I) is flowing from terminals A to B through a passive element having impedance of $Z \Omega$. It is the ratio of voltage (V) across that element between terminals A & B and current (I).

In the above figure, the current (I) is flowing from terminals B to A through a passive element having



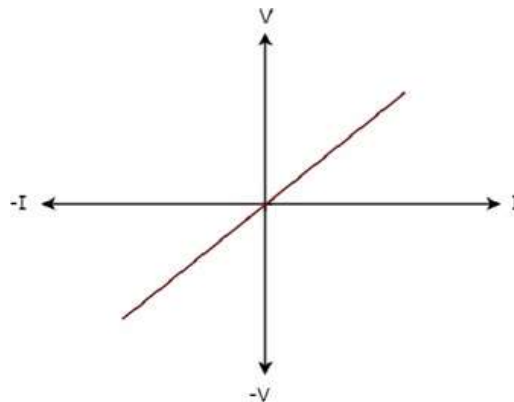
impedance of $Z \Omega$. That means the current ($-I$) is flowing from terminals A to B. In this case too, we will get the same impedance value, since both the current and voltage having negative signs with respect to terminals A & B.

Unilateral Elements are those that allow the current in only one direction. Hence, they offer different impedances in both directions.

We discussed the types of network elements in the previous chapter. Now, let us identify the nature of network elements from the V-I characteristics given in the following examples.

Example 1

The V-I characteristics of a network element is shown below.



Step 1 – Verifying the network element as linear or non-linear.

From the above figure, the V-I characteristics of a network element is a straight line passing through the origin. Hence, it is linear element.

Step 2 – Verifying the network element as active or passive.

The given V-I characteristics of a network element lies in the first and third quadrants.

- In the first quadrant, the values of both voltage (V) and current (I) are positive. So, the ratios of voltage (V) and current (I) gives positive impedance values.

- Similarly, in the third quadrant, the values of both voltage (V) and current (I) have negative values.

So, the ratios of voltage (V) and current (I) produce positive impedance values.

Since, the given V-I characteristics offer positive impedance values, the network element is a Passive element.

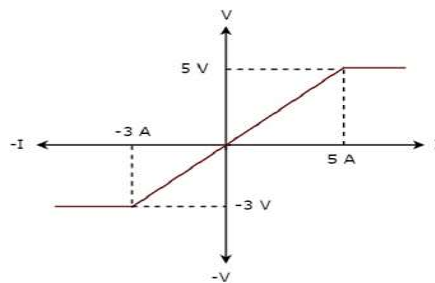
Step 3 – Verifying the network element as bilateral or unilateral.

For every point (I, V) on the characteristics, there exists a corresponding point (-I, -V) on the given characteristics. Hence, the network element is a Bilateral element.

Therefore, the given V-I characteristics show that the network element is a Linear, Passive, and Bilateral element.

Example 2

The V-I characteristics of a network element is shown below.



Step 1 – Verifying the network element as linear or non-linear.

From the above figure, the V-I characteristics of a network element is a straight line only between the points (-3A, -3V) and (5A, 5V). Beyond these points, the V-I characteristics are not following the linear relation. Hence, it is a Non-linear element.

Step 2 – Verifying the network element as active or passive.

The given V-I characteristics of a network element lies in the first and third quadrants. In these two quadrants, the ratios of voltage (V) and current (I) produce positive impedance values. Hence, the network element is a Passive element.

Step 3 – Verifying the network element as bilateral or unilateral.

Consider the point (5A, 5V) on the characteristics. The corresponding point (-5A, -3V) exists on the given characteristics instead of (-5A, -5V). Hence, the network element is a Unilateral element.

Therefore, the given V-I characteristics show that the network element is a Non-linear, Passive, and Unilateral element. The circuits containing them are called unilateral circuits.

Lumped and Distributed Elements

Lumped elements are those elements which are very small in size & in which simultaneous actions takes place. Typical lumped elements are capacitors, resistors, inductors.

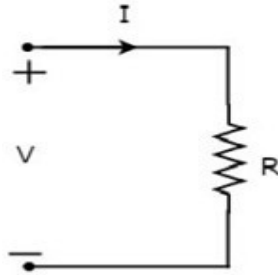
Distributed elements are those which are not electrically separable for analytical purposes.

For example a transmission line has distributed parameters along its length and may extend for hundreds of miles.

R-L-C Parameters

Resistor

The main functionality of Resistor is either opposes or restricts the flow of electric current. Hence, the resistors are used in order to limit the amount of current flow and / or dividing (sharing) voltage. Let the current flowing through the resistor is I amperes and the voltage across it is V volts. The symbol of resistor along with current, I and voltage, V are shown in the following figure.



According to Ohm's law, the voltage across resistor is the product of current flowing through it and the resistance of that resistor. Mathematically, it can be represented as

$$V = IR \quad \text{Equation 1}$$

$$\Rightarrow I = \frac{V}{R} \quad \text{Equation 2}$$

Where, **R** is the resistance of a resistor.

From Equation 2, we can conclude that the current flowing through the resistor is directly proportional to the applied voltage across resistor and inversely proportional to the resistance of resistor.

Power in an electric circuit element can be represented as

$$P = VI \quad \text{Equation 3}$$

Substitute, Equation 1 in Equation 3.

$$\begin{aligned} P &= (IR)I \\ \Rightarrow P &= I^2 R \end{aligned} \quad \text{Equation 4}$$

Substitute, Equation 2 in Equation 3.

$$\begin{aligned} P &= V\left(\frac{V}{R}\right) \\ \Rightarrow P &= \frac{V^2}{R} \end{aligned} \quad \text{Equation 5}$$

So, we can calculate the amount of power dissipated in the resistor by using one of the formulae mentioned in Equations 3 to 5.

Inductor

In general, inductors will have number of turns. Hence, they produce magnetic flux when current flows through it. So, the amount of total magnetic flux produced by an inductor depends on the current, I flowing through it and they have linear relationship.

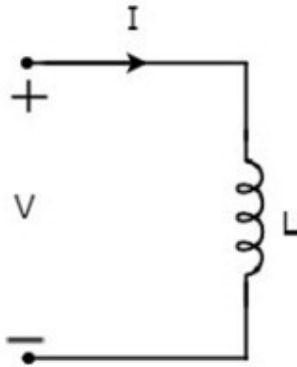
Mathematically, it can be written as

$$\Psi \propto I$$
$$\Rightarrow \Psi = LI$$

Where,

- Ψ is the total magnetic flux
- L is the inductance of an inductor

Let the current flowing through the inductor is I amperes and the voltage across it is V volts. The symbol of inductor along with current I and voltage V are shown in the following figure.



According to Faraday's law, the voltage across the inductor can be written as

$$V = \frac{d\Psi}{dt}$$

Substitute $\Psi = LI$ in the above equation.

$$V = \frac{d(LI)}{dt}$$
$$\Rightarrow V = L \frac{dI}{dt}$$
$$\Rightarrow I = \frac{1}{L} \int V dt$$

From the above equations, we can conclude that there exists a linear relationship between voltage across inductor and current flowing through it.

We know that power in an electric circuit element can be represented as

$$P = VI$$

Substitute $V = L \frac{dI}{dt}$ in the above equation.

$$P = (L \frac{dI}{dt})I$$

$$\Rightarrow P = LI \frac{dI}{dt}$$

By integrating the above equation, we will get the energy stored in an inductor as

$$W = \frac{1}{2} LI^2$$

So, the inductor stores the energy in the form of magnetic field.

Capacitor

In general, a capacitor has two conducting plates, separated by a dielectric medium. If positive voltage is applied across the capacitor, then it stores positive charge. Similarly, if negative voltage is applied across the capacitor, then it stores negative charge.

So, the amount of charge stored in the capacitor depends on the applied voltage V across it and they have linear relationship. Mathematically, it can be written as

$$Q \propto V$$

$$\Rightarrow Q = CV$$

Where,

- Q is the charge stored in the capacitor.
- C is the capacitance of a capacitor.

Let the current flowing through the capacitor is I amperes and the voltage across it is V volts. The symbol of capacitor along with current I and voltage V are shown in the following figure.

We know that the **current** is nothing but the **time rate of flow of charge**. Mathematically, it can be

I

$$I = \frac{dQ}{dt}$$

Substitute $Q = CV$ in the above equation.

$$I = \frac{d(CV)}{dt}$$

$$\Rightarrow I = C \frac{dV}{dt}$$

$$\Rightarrow V = \frac{1}{C} \int Idt$$

represented as

From the above equations, we can conclude that there exists a linear relationship between voltage across capacitor and current flowing through it.

We know that power in an electric circuit element can be represented as

$$P = VI$$

Substitute $I = C \frac{dV}{dt}$ in the above equation.

$$P = V(C \frac{dV}{dt})$$

$$\Rightarrow P = CV \frac{dV}{dt}$$

By integrating the above equation, we will get the **energy** stored in the capacitor as

$$W = \frac{1}{2} CV^2$$

So, the capacitor stores the energy in the form of electric field.

Types of Sources

Active Elements are the network elements that deliver power to other elements present in an electric circuit. So, active elements are also called as sources of voltage or current type. We can classify these sources into the following two categories –

- Independent Sources
- Dependent Sources

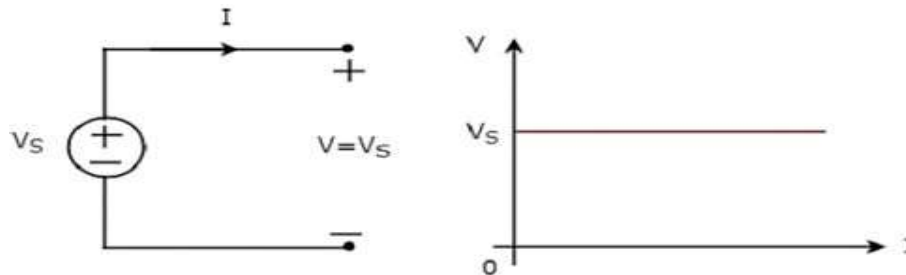
Independent Sources

As the name suggests, independent sources produce fixed values of voltage or current and these are not dependent on any other parameter. Independent sources can be further divided into the following two categories –

- Independent Voltage Sources
- Independent Current Sources

Independent Voltage Sources

An independent voltage source produces a constant voltage across its two terminals. This voltage is independent of the amount of current that is flowing through the two terminals of voltage source. Independent ideal voltage source and its V-I characteristics are shown in the following figure.

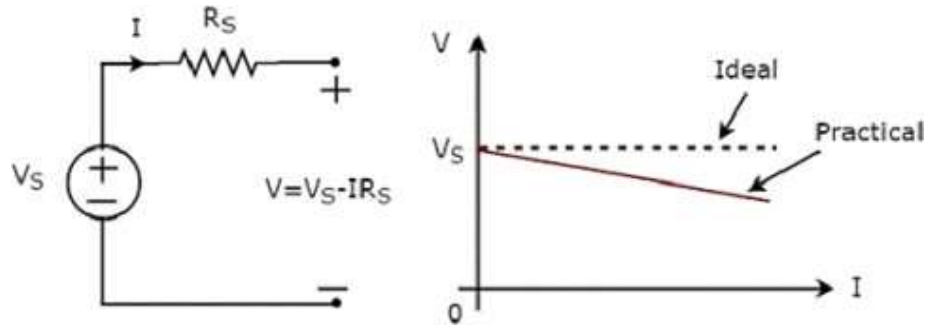


The V-I characteristics of an independent ideal voltage source is a constant line, which is always equal to the source voltage (V_S) irrespective of the current value (I). So, the internal resistance of an independent ideal voltage source is zero Ohms.

Hence, the independent ideal voltage sources do not exist practically, because there will be some internal resistance.

Independent practical voltage source and its V-I characteristics are shown in the following figure.

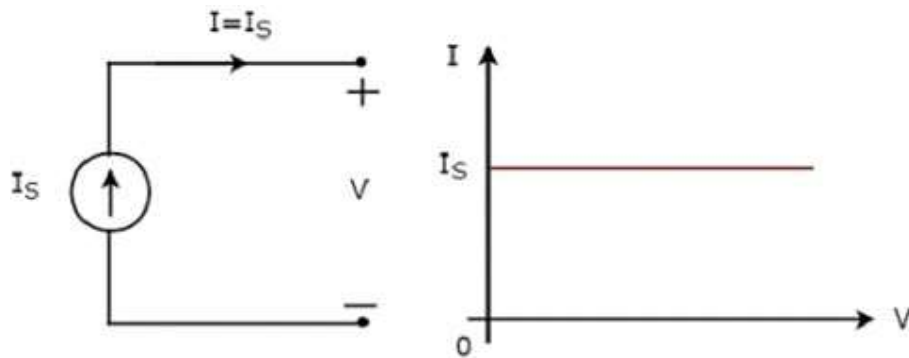
There is a deviation in the V-I characteristics of an independent practical voltage source from the V-I



characteristics of an independent ideal voltage source. This is due to the voltage drop across the internal resistance (R_S) of an independent practical voltage source.

Independent Current Sources

An independent current source produces a constant current. This current is independent of the voltage across its two terminals. Independent ideal current source and its V-I characteristics are shown in the following figure.

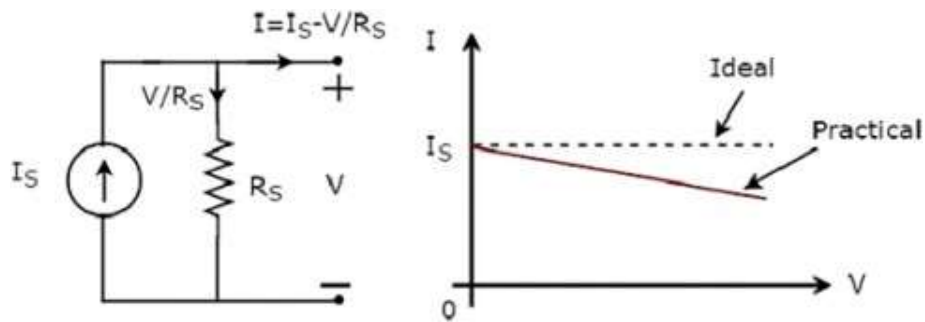


The V-I characteristics of an independent ideal current source is a constant line, which is always equal to the source current (I_S) irrespective of the voltage value (V). So, the internal resistance of an independent ideal current source is infinite ohms.

Hence, the independent ideal current sources do not exist practically, because there will be some internal resistance.

Independent practical current source and its V-I characteristics are shown in the following figure.

There is a deviation in the V-I characteristics of an independent practical current source from the V-I



characteristics of an independent ideal current source. This is due to the amount of current flows through the internal shunt resistance (R_S) of an independent practical current source.

Dependent Sources

As the name suggests, dependent sources produce the amount of voltage or current that is dependent on some other voltage or current. Dependent sources are also called as controlled sources. Dependent sources can be further divided into the following two categories –

- Dependent Voltage Sources
- Dependent Current Sources

Dependent Voltage Sources

A dependent voltage source produces a voltage across its two terminals. The amount of this voltage is dependent on some other voltage or current. Hence, dependent voltage sources can be further classified into the following two categories –

- Voltage Dependent Voltage Source (VDVS)
- Current Dependent Voltage Source (CDVS)

Dependent voltage sources are represented with the signs '+' and '-' inside a diamond shape. The magnitude of the voltage source can be represented outside the diamond shape.

Dependent Current Sources

A dependent current source produces a current. The amount of this current is dependent on some other voltage or current. Hence, dependent current sources can be further classified into the following two categories –

- Voltage Dependent Current Source (VDCS)
- Current Dependent Current Source (CDCS)

Dependent current sources are represented with an arrow inside a diamond shape. The magnitude of the current source can be represented outside the diamond shape. We can observe these dependent or controlled sources in equivalent models of transistors.

Source Transformation Technique

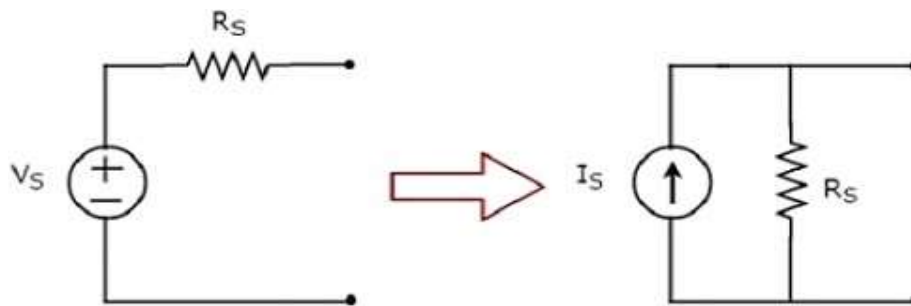
We know that there are two practical sources, namely, voltage source and current source. We can transform (convert) one source into the other based on the requirement, while solving network problems.

The technique of transforming one source into the other is called as source transformation technique. Following are the two possible source transformations –

- Practical voltage source into a practical current source
- Practical current source into a practical voltage source

Practical voltage source into a practical current source

The transformation of practical voltage source into a practical current source is shown in the following figure



Practical voltage source consists of a voltage source (V_S) in series with a resistor (R_S). This can be converted into a practical current source as shown in the figure. It consists of a current source (I_S) in parallel with a resistor (R_S).

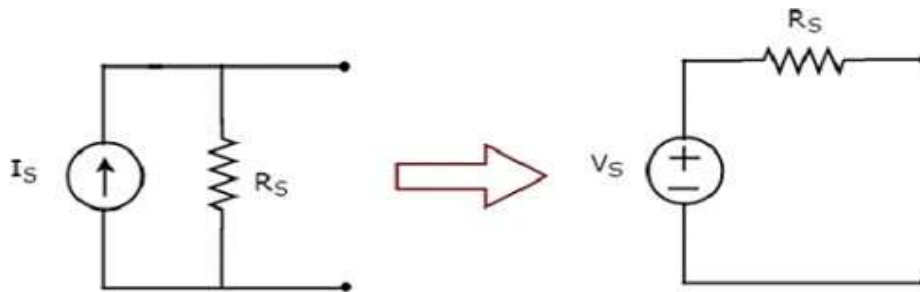
The value of I_S will be equal to the ratio of V_S and R_S . Mathematically, it can be represented as

$$I_S = \frac{V_S}{R_S}$$

Practical current source into a practical voltage source

The transformation of practical current source into a practical voltage source is shown in the following figure.

Practical current source consists of a current source (I_S) in parallel with a resistor (R_S). This can be



converted into a practical voltage source as shown in the figure. It consists of a voltage source (V_S) in series with a resistor (R_S).

The value of V_S will be equal to the product of I_S and R_S . Mathematically, it can be represented as

$$V_S = I_S R_S$$

In this chapter, we will discuss in detail about the passive elements such as Resistor, Inductor, and Capacitor. Let us start with Resistors.

Kirchhoff's Laws

Network elements can be either of active or passive type. Any electrical circuit or network contains one of these two types of network elements or a combination of both.

Now, let us discuss about the following two laws, which are popularly known as Kirchhoff's laws.

- Kirchhoff's Current Law
- Kirchhoff's Voltage Law

Kirchhoff's Current Law

Kirchhoff's Current Law (KCL) states that the algebraic sum of currents leaving (or entering) a node is equal to zero.

A Node is a point where two or more circuit elements are connected to it. If only two circuit elements are connected to a node, then it is said to be simple node. If three or more circuit elements are connected to a node, then it is said to be Principal Node.

Mathematically, KCL can be represented as

$$\sum_{m=1}^M I_m = 0$$

Where,

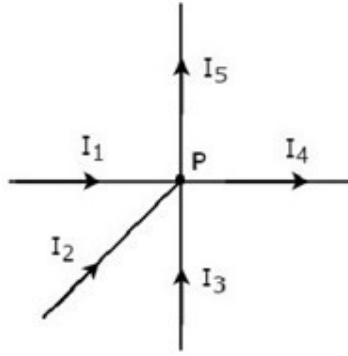
- I_m is the m^{th} branch current leaving the node.

- M is the number of branches that are connected to a node.

The above statement of KCL can also be expressed as "the algebraic sum of currents entering a node is equal to the algebraic sum of currents leaving a node". Let us verify this statement through the following example.

Example

Write KCL equation at node P of the following figure.



- In the above figure, the branch currents I_1 , I_2 and I_3 are entering at node P. So, consider negative signs for these three currents.
- In the above figure, the branch currents I_4 and I_5 are leaving from node P. So, consider positive signs for these two currents.

The KCL equation at node P will be

$$-I_1 - I_2 - I_3 + I_4 + I_5 = 0$$

$$\Rightarrow I_1 + I_2 + I_3 = I_4 + I_5$$

In the above equation, the left-hand side represents the sum of entering currents, whereas the right-hand side represents the sum of leaving currents.

In this tutorial, we will consider positive sign when the current leaves a node and negative sign when it enters a node. Similarly, you can consider negative sign when the current leaves a node and positive sign when it enters a node. In both cases, the result will be same.

Note – KCL is independent of the nature of network elements that are connected to a node.

Kirchhoff's Voltage Law

Kirchhoff's Voltage Law (KVL) states that the algebraic sum of voltages around a loop or mesh is equal to zero.

A Loop is a path that terminates at the same node where it started from. In contrast, a Mesh is a loop that doesn't contain any other loops inside it.

Mathematically, KVL can be represented as

$$\sum_{n=1}^N V_n = 0$$

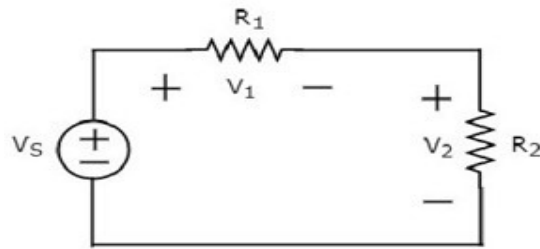
Where,

- V_n is the n^{th} element's voltage in a loop (mesh).
- N is the number of network elements in the loop (mesh).

The above statement of KVL can also be expressed as "the algebraic sum of voltage sources is equal to the algebraic sum of voltage drops that are present in a loop." Let us verify this statement with the help of the following example.

Example

Write KVL equation around the loop of the following circuit.



The above circuit diagram consists of a voltage source, V_S in series with two resistors R_1 and R_2 . The voltage drops across the resistors R_1 and R_2 are V_1 and V_2 respectively.

Apply KVL around the loop.

$$V_S - V_1 - V_2 = 0$$

$$\Rightarrow V_S = V_1 + V_2$$

In the above equation, the left-hand side term represents single voltage source V_S . Whereas, the right-hand side represents the sum of voltage drops. In this example, we considered only one voltage source. That's why the left-hand side contains only one term. If we consider multiple voltage sources, then the left side contains sum of voltage sources.

In this tutorial, we consider the sign of each element's voltage as the polarity of the second terminal that is present while travelling around the loop. Similarly, you can consider the sign of each voltage as the polarity of the first terminal that is present while travelling around the loop. In both cases, the result will be same.

Note – KVL is independent of the nature of network elements that are present in a loop.

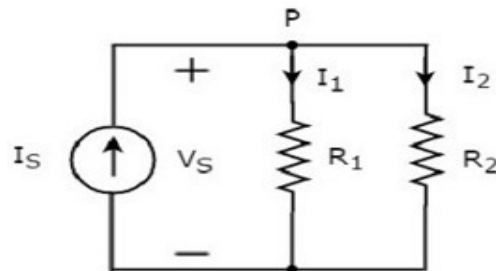
In this chapter, let us discuss about the following two division principles of electrical quantities.

- Current Division Principle
- Voltage Division Principle

Current Division Principle

When two or more passive elements are connected in parallel, the amount of current that flows through each element gets divided(shared) among themselves from the current that is entering the node.

Consider the following circuit diagram.



The above circuit diagram consists of an input current source I_S in parallel with two resistors R_1 and R_2 . The voltage across each element is V_S . The currents flowing through the resistors R_1 and R_2 are I_1 and I_2 respectively.

The KCL equation at node P will be

$$I_S = I_1 + I_2$$

- Substitute $I_1 = \frac{V_S}{R_1}$ and $I_2 = \frac{V_S}{R_2}$ in the above equation.

$$I_S = \frac{V_S}{R_1} + \frac{V_S}{R_2} = V_S \left(\frac{R_2 + R_1}{R_1 R_2} \right)$$

$$\Rightarrow V_S = I_S \left(\frac{R_1 R_2}{R_1 + R_2} \right)$$

- Substitute the value of V_S in $I_1 = \frac{V_S}{R_1}$.

$$I_1 = \frac{I_S}{R_1} \left(\frac{R_1 R_2}{R_1 + R_2} \right)$$

$$\Rightarrow I_1 = I_S \left(\frac{R_2}{R_1 + R_2} \right)$$

- Substitute the value of V_S in $I_2 = \frac{V_S}{R_2}$.

$$I_2 = \frac{I_S}{R_2} \left(\frac{R_1 R_2}{R_1 + R_2} \right)$$

$$\Rightarrow I_2 = I_S \left(\frac{R_1}{R_1 + R_2} \right)$$

From equations of I_1 and I_2 , we can generalize that the current flowing through any passive element can be found by using the following formula.

$$I_N = I_S \left(\frac{Z_1 \parallel Z_2 \parallel \dots \parallel Z_{N-1}}{Z_1 + Z_2 + \dots + Z_N} \right)$$

This is known as current division principle and it is applicable, when two or more passive elements are connected in parallel and only one current enters the node.

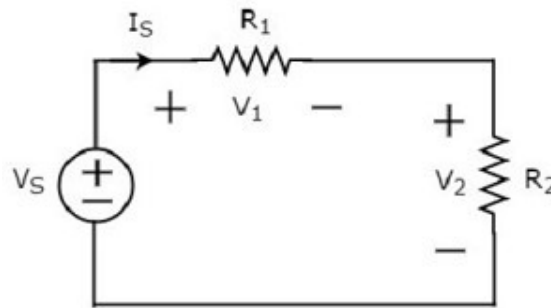
Where,

- I_N is the current flowing through the passive element of N^{th} branch.
- I_S is the input current, which enters the node.
- $Z_1, Z_2, \dots, Z_N$ are the impedances of 1^{st} branch, 2^{nd} branch, $\dots$, N^{th} branch respectively.

Voltage Division Principle

When two or more passive elements are connected in series, the amount of voltage present across each element gets divided (shared) among themselves from the voltage that is available across that entire combination.

Consider the following circuit diagram.



The above circuit diagram consists of a voltage source, V_S in series with two resistors R_1 and R_2 . The current flowing through these elements is I_S . The voltage drops across the resistors R_1 and R_2 are V_1 and V_2 respectively.

The KVL equation around the loop will be

$$V_S = V_1 + V_2$$

- Substitute $V_1 = I_S R_1$ and $V_2 = I_S R_2$ in the above equation

$$V_S = I_S R_1 + I_S R_2 = I_S (R_1 + R_2)$$

$$I_S = \frac{V_S}{R_1 + R_2}$$

- Substitute the value of I_S in $V_1 = I_S R_1$.

$$V_1 = \left(\frac{V_S}{R_1 + R_2} \right) R_1$$

$$\Rightarrow V_1 = V_S \left(\frac{R_1}{R_1 + R_2} \right)$$

- Substitute the value of I_S in $V_2 = I_S R_2$.

$$V_2 = \left(\frac{V_S}{R_1 + R_2} \right) R_2$$

$$\Rightarrow V_2 = V_S \left(\frac{R_2}{R_1 + R_2} \right)$$

From equations of V_1 and V_2 , we can generalize that the voltage across any passive element can be found by using the following formula.

$$V_N = V_S \left(\frac{Z_N}{Z_1 + Z_2 + \dots + Z_N} \right)$$

This is known as voltage division principle and it is applicable, when two or more passive elements are connected in series and only one voltage available across the entire combination.

Where,

- V_N is the voltage across N^{th} passive element.
- V_S is the input voltage, which is present across the entire combination of series passive elements.
- $Z_1, Z_2, \dots, Z_N$ are the impedances of 1^{st} passive element, 2^{nd} passive element, ..., N^{th} passive element respectively.

UNIT-II

NETWORK ANALYSIS

- Network Reduction Techniques
- Series and Parallel connection of Resistive Networks
- Star-to-Delta and Delta-to-Star Transformations for Resistive Networks
- Mesh Analysis
- Network Theorems: Thevenin's Theorem
- Norton's Theorem
- Superposition Theorem
- Problems

Network Reduction Techniques:

There are two basic methods that are used for solving any electrical network: Nodal analysis and Mesh analysis. In this chapter, let us discuss about the Mesh analysis method.

Series and parallel connections of resistive networks:

If a circuit consists of two or more similar passive elements and are connected in exclusively of series type or parallel type, then we can replace them with a single equivalent passive element. Hence, this circuit is called as an equivalent circuit.

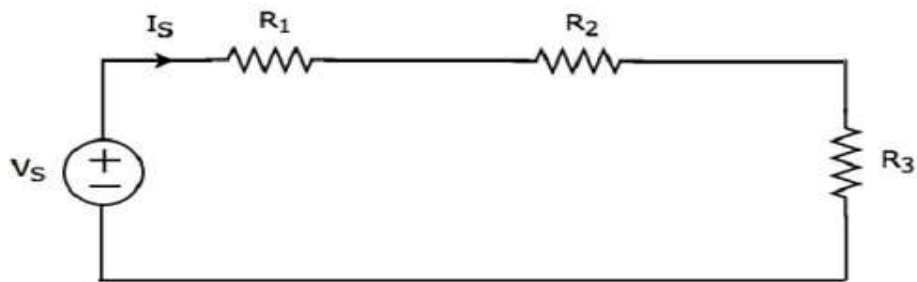
In this chapter, let us discuss about the following two equivalent circuits.

- Series Equivalent Circuit
- Parallel Equivalent Circuit

Series Equivalent Circuit

If similar passive elements are connected in series, then the same current will flow through all these elements. But, the voltage gets divided across each element.

Consider the following circuit diagram.



It has a single voltage source (V_S) and three resistors having resistances of R_1 , R_2 and R_3 . All these elements are connected in series. The current I_S flows through all these elements.

The above circuit has only one mesh. The KVL equation around this mesh is

$$V_S = V_1 + V_2 + V_3$$

Substitute $V_1 = I_S R_1$, $V_2 = I_S R_2$ and $V_3 = I_S R_3$ in the above equation.

$$V_S = I_S R_1 + I_S R_2 + I_S R_3$$

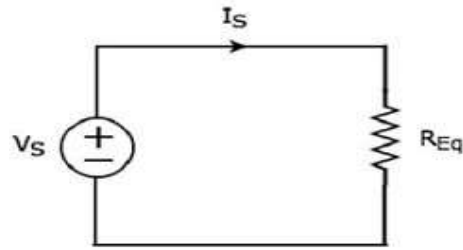
$$\Rightarrow V_S = I_S (R_1 + R_2 + R_3)$$

The above equation is in the form of $V_S = I_S R_{Eq}$ where,

$$R_{Eq} = R_1 + R_2 + R_3$$

The equivalent circuit diagram of the given circuit is shown in the following figure.

That means, if multiple resistors are connected in series, then we can replace them with an equivalent



resistor. The resistance of this equivalent resistor is equal to sum of the resistances of all those multiple resistors.

Note 1 – If ‘N’ inductors having inductances of $L_1, L_2, \dots, L_N$ are connected in series, then the equivalent inductance will be

$$L_{Eq} = L_1 + L_2 + \dots + L_N$$

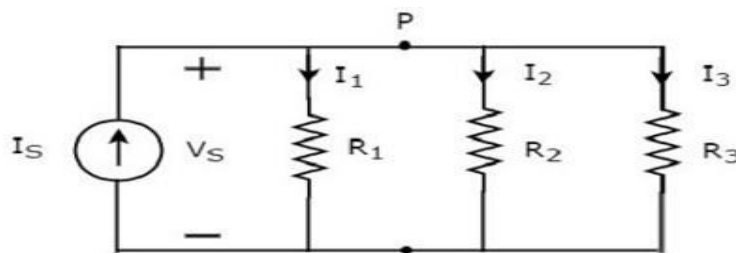
Note 2 – If ‘N’ capacitors having capacitances of $C_1, C_2, \dots, C_N$ are connected in series, then the equivalent capacitance will be

$$\frac{1}{C_{Eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$

Parallel Equivalent Circuit

If similar passive elements are connected in parallel, then the same voltage will be maintained across each element. But, the current flowing through each element gets divided.

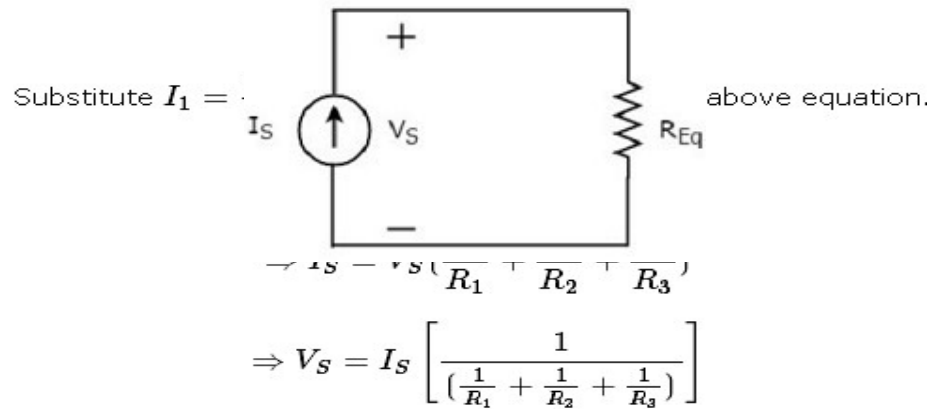
Consider the following circuit diagram.



It has a single current source (I_S) and three resistors having resistances of $R_1, R_2,$ and R_3 . All these elements are connected in parallel. The voltage (V_S) is available across all these elements.

The above circuit has only one principal node (P) except the Ground node. The KCL equation at this principal node (P) is

The equivalent circuit diagram of the given circuit is shown in the following figure.



The above equation is in the form of $V_S = I_S R_{Eq}$ where,

$$R_{Eq} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)}$$

$$\frac{1}{R_{Eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

That means, if multiple resistors are connected in parallel, then we can replace them with an equivalent resistor. The resistance of this equivalent resistor is equal to the reciprocal of sum of reciprocal of each resistance of all those multiple resistors.

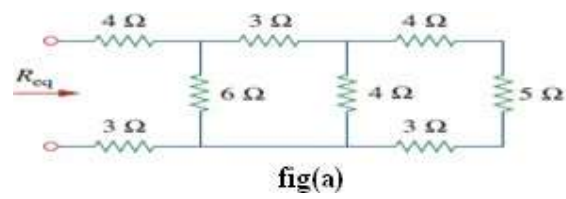
Note 1 – If ‘N’ inductors having inductances of $L_1, L_2, \dots, L_N$ are connected in parallel, then the equivalent inductance will be

$$\frac{1}{L_{Eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$$

Note 2 – If ‘N’ capacitors having capacitances of $C_1, C_2, \dots, C_N$ are connected in parallel, then the equivalent capacitance will be

$$C_{Eq} = C_1 + C_2 + \dots + C_N$$

1) Find the Req for the circuit shown in below figure.



Star-to-Delta and Delta-to-Star Transformations for Resistive Networks:

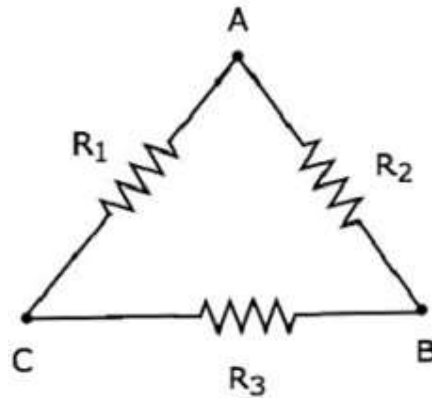
Delta to Star Transformation

In the previous chapter, we discussed an example problem related equivalent resistance. There, we calculated the equivalent resistance between the terminals A & B of the given electrical network easily. Because, in every step, we got the combination of resistors that are connected in either series form or parallel form.

However, in some situations, it is difficult to simplify the network by following the previous approach. For example, the resistors connected in either delta (δ) form or star form. In such situations, we have to convert the network of one form to the other in order to simplify it further by using series combination or parallel combination. In this chapter, let us discuss about the Delta to Star Conversion.

Delta Network

Consider the following delta network as shown in the following figure.



The following equations represent the equivalent resistance between two terminals of delta network, when the third terminal is kept open.

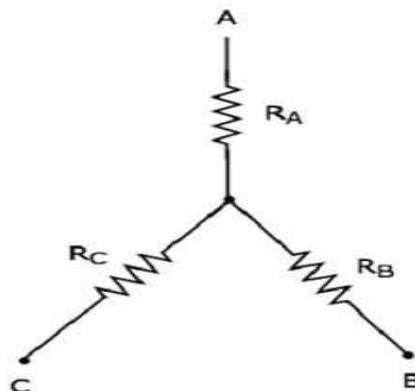
$$R_{AB} = \frac{(R_1 + R_3)R_2}{R_1 + R_2 + R_3}$$

$$R_{BC} = \frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3}$$

$$R_{CA} = \frac{(R_2 + R_3)R_1}{R_1 + R_2 + R_3}$$

Star Network

The following figure shows the equivalent star network corresponding to the above delta network.



The following equations represent the equivalent resistance between two terminals of star network, when the third terminal is kept open.

$$R_{AB} = R_A + R_B$$

$$R_{BC} = R_B + R_C$$

$$R_{CA} = R_C + R_A$$

Star Network Resistances in terms of Delta Network Resistances

We will get the following equations by equating the right-hand side terms of the above equations for which the left-hand side terms are same.

$$R_A + R_B = \frac{(R_1 + R_3)R_2}{R_1 + R_2 + R_3} \quad \text{Equation 1}$$

$$R_B + R_C = \frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3} \quad \text{Equation 2}$$

$$R_C + R_A = \frac{(R_2 + R_3)R_1}{R_1 + R_2 + R_3} \quad \text{Equation 3}$$

By adding the above three equations, we will get

$$\begin{aligned} 2(R_A + R_B + R_C) &= \frac{2(R_1 R_2 + R_2 R_3 + R_3 R_1)}{R_1 + R_2 + R_3} \\ \Rightarrow R_A + R_B + R_C &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 + R_2 + R_3} \quad \text{Equation 4} \end{aligned}$$

Subtract Equation 2 from Equation 4.

$$R_A + R_B + R_C - (R_B + R_C) = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 + R_2 + R_3} - \frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3}$$

$$R_A = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

By subtracting Equation 3 from Equation 4, we will get

$$R_B = \frac{R_2 R_3}{R_1 + R_2 + R_3}$$

By subtracting Equation 1 from Equation 4, we will get

$$R_C = \frac{R_3 R_1}{R_1 + R_2 + R_3}$$

By using the above relations, we can find the resistances of star network from the resistances of delta network. In this way, we can convert a delta network into a star network.

Star to Delta Transformation

In the previous chapter, we discussed about the conversion of delta network into an equivalent star network. Now, let us discuss about the conversion of star network into an equivalent delta network. This conversion is called as Star to Delta Conversion.

In the previous chapter, we got the resistances of star network from delta network as

$$R_A = \frac{R_1 R_2}{R_1 + R_2 + R_3} \quad \text{Equation 1}$$

$$R_B = \frac{R_2 R_3}{R_1 + R_2 + R_3} \quad \text{Equation 2}$$

$$R_C = \frac{R_3 R_1}{R_1 + R_2 + R_3} \quad \text{Equation 3}$$

Delta Network Resistances in terms of Star Network Resistances

Let us manipulate the above equations in order to get the resistances of delta network in terms of resistances of star network.

- Multiply each set of two equations and then add.

By using the above relations, we can find the resistances of delta network from the resistances of star

$$R_A R_B + R_B R_C + R_C R_A = \frac{R_1 R_2^2 R_3 + R_2 R_3^2 R_1 + R_3 R_1^2 R_2}{(R_1 + R_2 + R_3)^2}$$

$$\Rightarrow R_A R_B + R_B R_C + R_C R_A = \frac{R_1 R_2 R_3 (R_1 + R_2 + R_3)}{(R_1 + R_2 + R_3)^2}$$

$$\Rightarrow R_A R_B + R_B R_C + R_C R_A = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3}$$

Equation 4

- ▣ By dividing Equation 4 with Equation 2, we will get

$$\frac{R_A R_B + R_B R_C + R_C R_A}{R_B} = R_1$$

$$\Rightarrow R_1 = R_C + R_A + \frac{R_C R_A}{R_B}$$

- ▣ By dividing Equation 4 with Equation 3, we will get

$$R_2 = R_A + R_B + \frac{R_A R_B}{R_C}$$

- ▣ By dividing Equation 4 with Equation 1, we will get

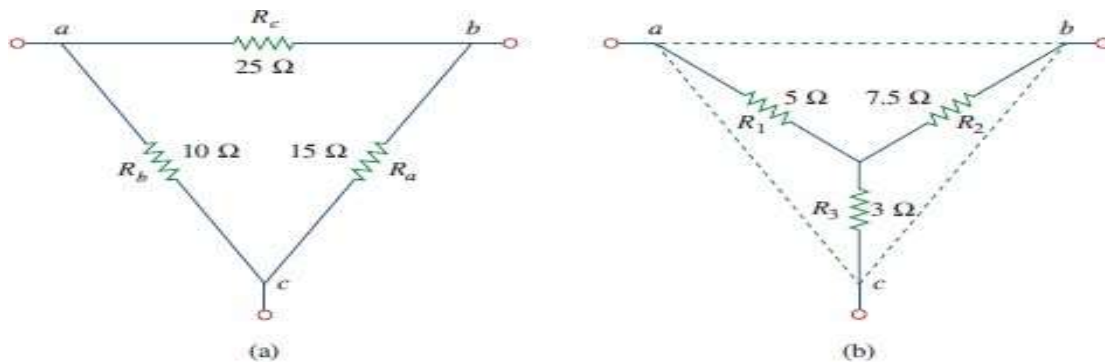
$$R_3 = R_B + R_C + \frac{R_B R_C}{R_A}$$

network. In this way, we can convert star network into delta network.

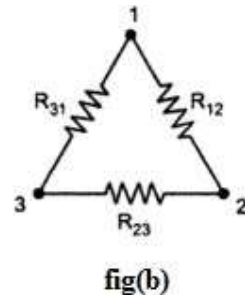
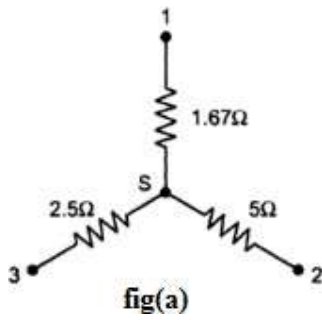
Example problems:

- 1) Convert the Delta network in Fig.(a) to an equivalent star network

Solution:



2) Convert the star network in fig(a) to delta network



Solution: The equivalent delta for the given star is shown in fig(b), where

Mesh Analysis:

Mesh analysis provides general procedure for analyzing circuits using mesh currents as the circuit variables. Mesh Analysis is applicable only for planar networks. It is preferably useful for the circuits that have many loops. This analysis is done by using KVL and Ohm's law.

In Mesh analysis, we will consider the currents flowing through each mesh. Hence, Mesh analysis is also called as Mesh-current method.

A branch is a path that joins two nodes and it contains a circuit element. If a branch belongs to only one mesh, then the branch current will be equal to mesh current.

If a branch is common to two meshes, then the branch current will be equal to the sum (or difference) of two mesh currents, when they are in same (or opposite) direction.

Procedure of Mesh Analysis

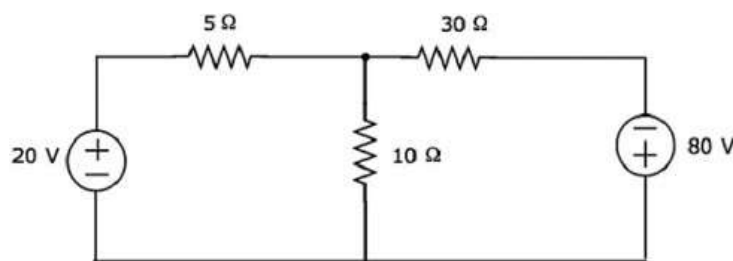
Follow these steps while solving any electrical network or circuit using Mesh analysis.

- **Step 1** – Identify the meshes and label the mesh currents in either clockwise or anti-clockwise direction.
- **Step 2** – Observe the amount of current that flows through each element in terms of mesh currents.
- **Step 3** – Write mesh equations to all meshes. Mesh equation is obtained by applying KVL first and then Ohm's law.
- **Step 4** – Solve the mesh equations obtained in Step 3 in order to get the mesh currents.

Now, we can find the current flowing through any element and the voltage across any element that is present in the given network by using mesh currents.

Example

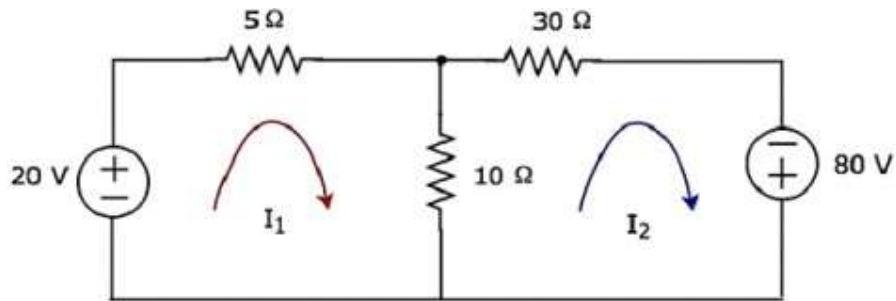
Find the voltage across $30\ \Omega$ resistor using Mesh analysis.



Step 1 – There are two meshes in the above circuit. The mesh currents I_1 and I_2 are considered in clockwise direction. These mesh currents are shown in the following figure.

Step 2 – The mesh current I_1 flows through 20 V voltage source and 5 Ω resistor. Similarly, the mesh current I_2 flows through 30 Ω resistor and -80 V voltage source. But, the difference of two mesh currents, I_1 and I_2 , flows through 10 Ω resistor, since it is the common branch of two meshes.

Step 3 – In this case, we will get two mesh equations since there are two meshes in the given circuit. When we write the mesh equations, assume the mesh current of that particular mesh as greater than all other mesh currents of the circuit. The mesh equation of first mesh is



Step 4 – Finding mesh currents I_1 and I_2 by solving Equation 1 and Equation 2.

The left-hand side terms of Equation 1 and Equation 2 are the same. Hence, equate the right-hand side terms of Equation 1 and Equation 2 in order to find the value of I_1 .

Therefore, the voltage across $30\ \Omega$ resistor of the given circuit is 84 V .

$$6I_1 - 8 = I_1 + 8$$

$$\Rightarrow 5I_1 = 16$$

$$\Rightarrow I_1 = \frac{16}{5}\text{ A}$$

Substitute I_1 value in Equation 2.

$$4I_2 = \frac{16}{5} + 8$$

$$\Rightarrow 4I_2 = \frac{56}{5}$$

$$\Rightarrow I_2 = \frac{14}{5}\text{ A}$$

So, we got the mesh currents I_1 and I_2 as $\frac{16}{5}\text{ A}$ and $\frac{14}{5}\text{ A}$ respectively.

Step 5 – The current flowing through $30\ \Omega$ resistor is nothing but the mesh current I_2 and it is equal to $\frac{14}{5}\text{ A}$. Now, we can find the voltage across $30\ \Omega$ resistor by using Ohm's law.

$$V_{30\Omega} = I_2 R$$

Substitute the values of I_2 and R in the above equation.

$$V_{30\Omega} = \left(\frac{14}{5}\right)30$$

$$\Rightarrow V_{30\Omega} = 84\text{V}$$

Note 1 – From the above example, we can conclude that we have to solve 'm' mesh equations, if the electric circuit is having 'm' meshes. That's why we can choose Mesh analysis when the number of meshes is less than the number of principal nodes (except the reference node) of any electrical circuit.

Note 2 – We can choose either Nodal analysis or Mesh analysis, when the number of meshes is equal to the number of principal nodes (except the reference node) in any electric circuit.

Network Theorems:

Introduction:

Any complicated network i.e. several sources, multiple resistors are present if the single element response is desired then use the network theorems. Network theorems are also can be termed as network reduction techniques. Each and every theorem got its importance of solving network. Let us see some important theorems with DC and AC excitation with detailed procedures.

Thevenin's Theorem and Norton's theorem (Introduction) :

Thevenin's Theorem and Norton's theorem are two important theorems in solving Network problems having many active and passive elements. Using these theorems the networks can be reduced to simple equivalent circuits with one active source and one element. In circuit analysis many a times the current through a branch is required to be found when it's value is changed with all other element values remaining same. In such cases finding out every time the branch current using the conventional mesh and node analysis methods is quite awkward and time consuming. But with the simple equivalent circuits (with one active source and one element) obtained using these two theorems the calculations become very simple. Thevenin's and Norton's theorems are dual theorems.

Thevenin's Theorem Statement:

Any linear, bilateral two terminal network consisting of sources and resistors(Impedance),can be replaced by an equivalent circuit consisting of a voltage source in series with a resistance (Impedance).The equivalent voltage source V_{Th} is the open circuit voltage looking into the terminals(with concerned branch element removed) and the equivalent resistance R_{Th} while all sources are replaced by their internal resistors at ideal condition i.e. voltage source is short circuit and current source is open circuit.

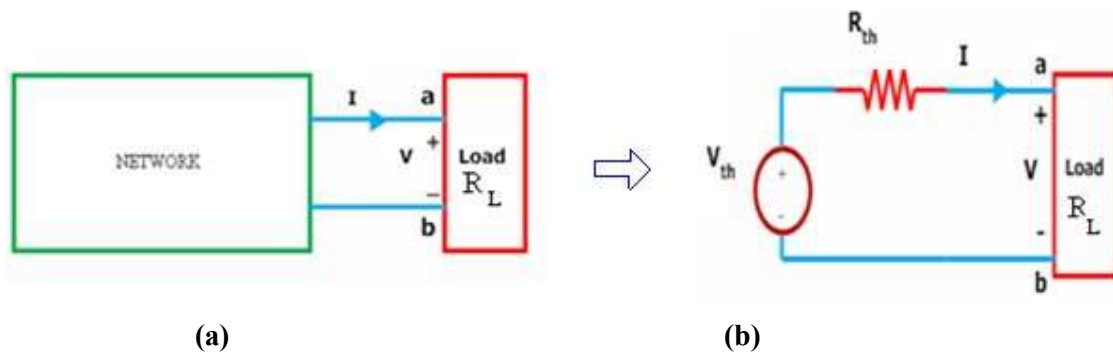


Figure (a) shows a simple block representation of a network with several active / passive elements with the load resistance R_L connected across the terminals 'a & b' and figure (b) shows the Thevenin's equivalent circuit with V_{Th} connected across R_{Th} & R_L .

Main steps to find out V_{Th} and R_{Th} :

1. The terminals of the branch/element through which the current is to be found out are marked as say a & b after removing the concerned branch/element
2. Open circuit voltage V_{OC} across these two terminals is found out using the conventional network mesh/node analysis methods and this would be V_{Th} .
3. Thevenin's resistance R_{Th} is found out by the method depending upon whether the network contains dependent sources or not.
 - a. With dependent sources: $R_{Th} = V_{oc} / I_{sc}$
 - b. Without dependent sources : R_{Th} = Equivalent resistance looking into the concerned terminals with all voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited)
4. Replace the network with V_{Th} in series with R_{Th} and the concerned branch resistance (or) load resistance across the load terminals (A&B) as shown in below fig.

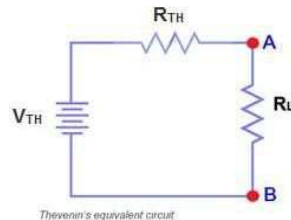
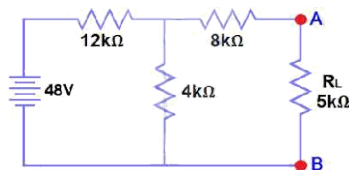


Fig.(a)

Example: Find V_{Th} , R_{Th} and the load current and load voltage flowing through R_L resistor as shown in fig. by using Thevenin's Theorem?



Solution:

The resistance R_L is removed and the terminals of the resistance R_L are marked as A & B as shown in the fig. (1)

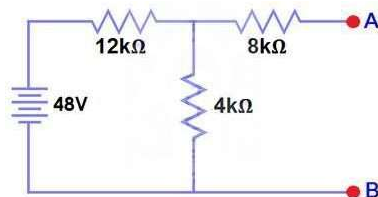


Fig.(1)

Calculate / measure the Open Circuit Voltage. This is the Thevenin Voltage (V_{TH}). We have already removed the load resistor from fig.(a), so the circuit became an open circuit as shown in fig (1). Now we have to calculate the Thevenin's Voltage. Since 3mA Current flows in both 12k Ω and 4k Ω resistors as this is a series circuit because current will not flow in the 8k Ω resistor as it is open. So 12V (3mA x 4k Ω) will appear across the 4k Ω resistor. We also know that current is not flowing through the 8k Ω resistor as it is open circuit, but the 8k Ω resistor is in parallel with 4k resistor. So the same voltage (i.e. 12V) will appear across the 8k Ω resistor as 4k Ω resistor. Therefore 12V will appear across the AB terminals.

So, $V_{TH} = 12V$

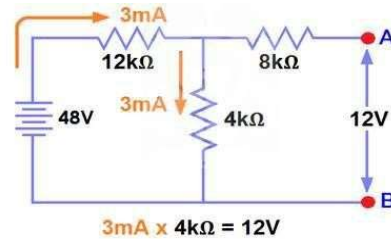
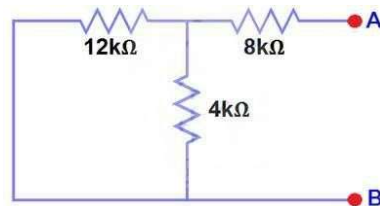


Fig (2)

All voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited) as shown in fig.(3)



Fig(3)

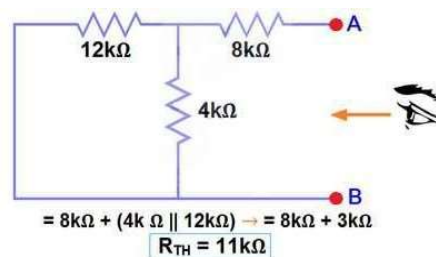
Calculate /measure the Open Circuit Resistance. This is the Thevenin's Resistance (R_{TH}) We have Reduced the 48V DC source to zero is equivalent to replace it with a short circuit as shown in figure (3) We can see that 8k Ω resistor is in series with a parallel connection of 4k Ω resistor and 12k Ω resistor. i.e.:

$8k\Omega + (4k\Omega \parallel 12k\Omega) \dots (\parallel = \text{in parallel with})$

$$R_{TH} = 8k\Omega + [(4k\Omega \times 12k\Omega) / (4k\Omega + 12k\Omega)]$$

$$R_{TH} = 8k\Omega + 3k\Omega$$

$$R_{TH} = 11k\Omega$$



Fig(4)

Connect the R_{TH} in series with Voltage Source V_{TH} and re-connect the load resistor across the load terminals(A&B) as shown in fig (5) i.e. Thevenin's circuit with load resistor. This is the Thevenin's equivalent circuit.

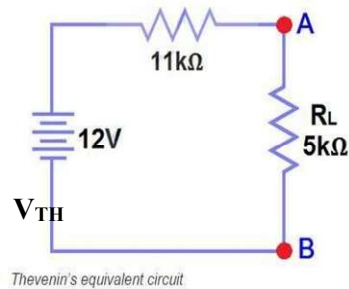


Fig (5)

Now apply Ohm's law and calculate the load current from fig 5.

$$I_L = V_{TH} / (R_{TH} + R_L) = 12V / (11k\Omega + 5k\Omega) = 12/16k\Omega$$

$$I_L = 0.75mA$$

$$\text{And } V_L = I_L \times R_L = 0.75mA \times 5k\Omega$$

$$V_L = 3.75V$$

Norton's Theorem Statement:

Any linear, bilateral two terminal network consisting of sources and resistors(Impedance),can be replaced by an equivalent circuit consisting of a current source in parallel with a resistance (Impedance),the current source being the short circuited current across the load terminals and the resistance being the internal resistance of the source network looking through the open circuited load terminals.

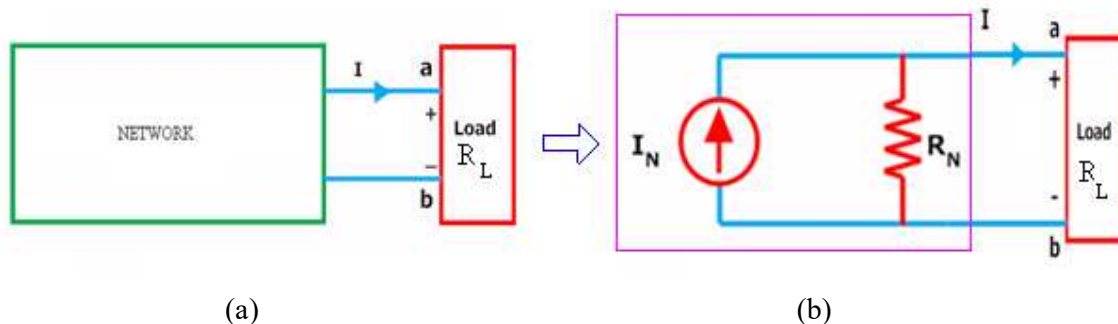
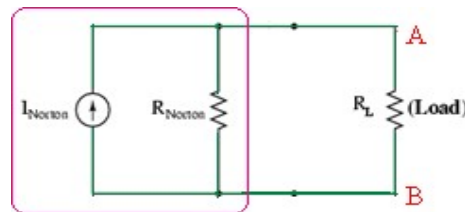


Figure (a) shows a simple block representation of a network with several active / passive elements with the load resistance R_L connected across the terminals 'a & b' and figure (b) shows the **Norton equivalent circuit** with I_N connected across R_N & R_L .

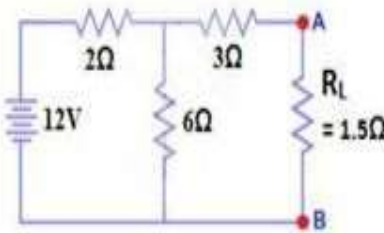
Main steps to find out I_N and R_N :

- The terminals of the branch/element through which the current is to be found out are marked as say **a & b** after removing the concerned branch/element.

- Open circuit voltage V_{OC} across these two terminals and I_{SC} through these two terminals are found out using the conventional network mesh/node analysis methods and they are same as what we obtained in Thevenin's equivalent circuit.
- Next **Norton resistance** R_N is found out depending upon whether the network contains dependent sources or not.
 - a) With dependent sources: $R_N = V_{oc} / I_{sc}$
 - b) Without dependent sources : $R_N =$ Equivalent resistance looking into the concerned terminals with all voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited)
- Replace the network with I_N in parallel with R_N and the concerned branch resistance across the load terminals(A&B) as shown in below fig

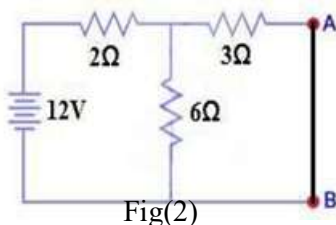


Example: Find the current through the resistance R_L (1.5Ω) of the circuit shown in the figure (a) below using Norton's equivalent circuit.



Fig(a)

Solution: To find out the Norton's equivalent ckt we have to find out $I_N = I_{sc}$, $R_N = V_{oc} / I_{sc}$. Short the 1.5Ω load resistor as shown in (Fig 2), and Calculate / measure the Short Circuit Current. This is the Norton Current (I_N).



Fig(2)

We have shorted the AB terminals to determine the Norton current, I_N . The 6Ω and 3Ω are then in parallel and this parallel combination of 6Ω and 3Ω are then in series with 2Ω . So the Total Resistance of the circuit to the Source is:-

$$2\Omega + (6\Omega \parallel 3\Omega) \dots (\parallel = \text{in parallel with})$$

$$R_T = 2\Omega + [(3\Omega \times 6\Omega) / (3\Omega + 6\Omega)]$$

$$R_T = 2\Omega + 2\Omega$$

$$R_T = 4\Omega$$

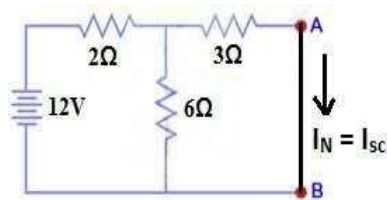
$$I_T = V / R_T$$

$$I_T = 12V / 4\Omega = 3A..$$

Now we have to find $I_{SC} = I_N$... Apply CDR... (Current Divider Rule)...

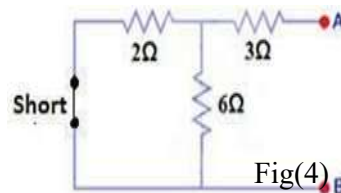
$$I_{SC} = I_N = 3A \times [(6\Omega / (3\Omega + 6\Omega))] = 2A.$$

$$I_{SC} = I_N = 2A.$$



Fig(3)

All voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited) and Open Load Resistor. as shown in fig.(4)



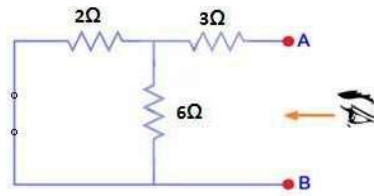
Calculate /measure the Open Circuit Resistance. This is the Norton Resistance (R_N) We have Reduced the 12V DC source to zero is equivalent to replace it with a short circuit as shown in fig(4), We can see that 3Ω resistor is in series with a parallel combination of 6Ω resistor and 2Ω resistor. i.e.:

$$3\Omega + (6\Omega \parallel 2\Omega) \dots (\parallel = \text{in parallel with})$$

$$R_N = 3\Omega + [(6\Omega \times 2\Omega) / (6\Omega + 2\Omega)]$$

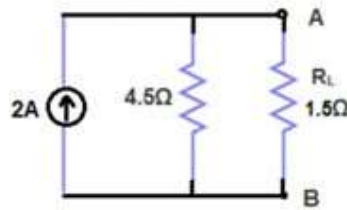
$$R_N = 3\Omega + 1.5\Omega$$

$$R_N = 4.5\Omega$$



Fig(5)

Connect the R_N in Parallel with Current Source I_N and re-connect the load resistor. This is shown in fig (6) i.e. Norton Equivalent circuit with load resistor.



Fig(6)

Now apply the Ohm's Law and calculate the load current through Load resistance across the terminals A&B. Load Current through Load Resistor is

$$I_L = I_N \times [R_N / (R_N + R_L)]$$

$$I_L = 2A \times (4.5\Omega / 4.5\Omega + 1.5k\Omega)$$

$$I_L = 1.5A \quad I_L = 1.5A$$

Superposition Theorem:

The principle of superposition helps us to analyze a linear circuit with more than one current or voltage sources sometimes it is easier to find out the voltage across or current in a branch of the circuit by considering the effect of one source at a time by replacing the other sources with their ideal internal resistances.

Superposition Theorem Statement:

Any linear, bilateral two terminal network consisting of more than one sources, The total current or voltage in any part of a network is equal to the algebraic sum of the currents or voltages in the required branch with each source acting individually while other sources are replaced by their ideal internal resistances. (i.e. Voltage sources by a short circuit and current sources by open circuit)

Steps to Apply Super position Principle:

1. Replace all independent sources with their internal resistances except one source. Find the output (voltage or current) due to that active source using nodal or mesh analysis.

2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

Example: By Using the superposition theorem find I in the circuit shown in figure?

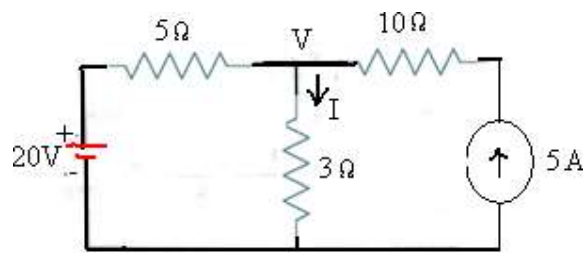


Fig.(a)

Solution: Applying the superposition theorem, the current I_2 in the resistance of $3\ \Omega$ due to the voltage source of 20V alone, with current source of 5A open circuited [as shown in the figure.1 below] is given by :

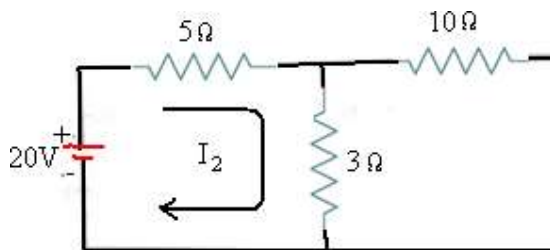


Fig.1

$$I_2 = 20/(5+3) = 2.5\text{A}$$

Similarly the current I_5 in the resistance of $3\ \Omega$ due to the current source of 5A alone with voltage source of 20V short circuited [as shown in the figure.2 below] is given by :

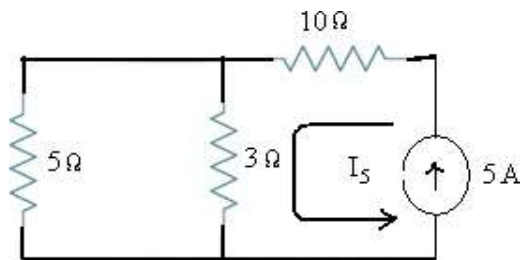


Fig.2

$$I_5 = 5 \times 5/(3+5) = 3.125\text{ A}$$

The total current passing through the resistance of $3\ \Omega$ is then $= I_2 + I_5 = 2.5 + 3.125 = \mathbf{5.625\text{ A}}$

Let us verify the solution using the basic nodal analysis referring to the node marked with V in fig.(a).Then we get :

$$\frac{V - 20}{5} + \frac{V}{3} = 5$$

$$3V - 60 + 5V = 15 \times 5$$

$$8V - 60 = 75$$

$$8V = 135$$

$$V = 16.875$$

The current I passing through the resistance of $3\Omega = V/3 = 16.875/3 = \mathbf{5.625 \text{ A}}$.